A-Level Physics Synoptic Assessment

Topic: Physics of the Microwave Oven

Synoptic Question: Physics of the Microwave Oven

Data required for this question:
Speed of electromagnetic waves in a vacuum, $c = 3.00 \times 10^8$ m s^{-1}
Elementary charge, $e = 1.60 \times 10^{-19}$ C
Mass of an electron, $m_e = 9.11 \times 10^{-31}$ kg
Specific heat capacity of water, $c_w = 4180$ J kg^{-1} K^{-1}

1. A standard domestic microwave oven operates at a frequency of 2.45 GHz. The microwaves are generated by a device called a magnetron. Inside the magnetron, electrons are emitted from a central hot cathode and accelerate towards a cylindrical anode. A strong, uniform magnetic field is applied parallel to the central axis, forcing the electrons into circular paths.

(a) The Magnetron (Electromagnetism)

(i) State the name of the process by which electrons are emitted from the hot cathode. [1]

(ii) An electron travels at a speed v perpendicular to a uniform magnetic field of flux density B. Show that the frequency of the electron's circular orbit (the cyclotron frequency) is independent of its speed v, and is given by the equation:

$$f = \frac{Be}{2\pi m_e}$$

[3]

(iii) Calculate the magnetic flux density B required to produce a cyclotron frequency equal to the operating frequency of the microwave oven (2.45 GHz). [2]

(b) Heating Mechanism (Electric Fields & Matter)
The microwaves generated by the magnetron are channeled into the oven cavity. Water molecules within the food are electric dipoles.

(i) Explain what is meant by an electric dipole. [1]

(ii) Explain how the oscillating electric field of the microwaves causes a temperature increase in food containing water. [3]

(c) Standing Waves (Wave Mechanics)
The microwave cavity acts as a resonator. The microwaves reflect off the metal walls, forming a three-dimensional stationary (standing) wave pattern.

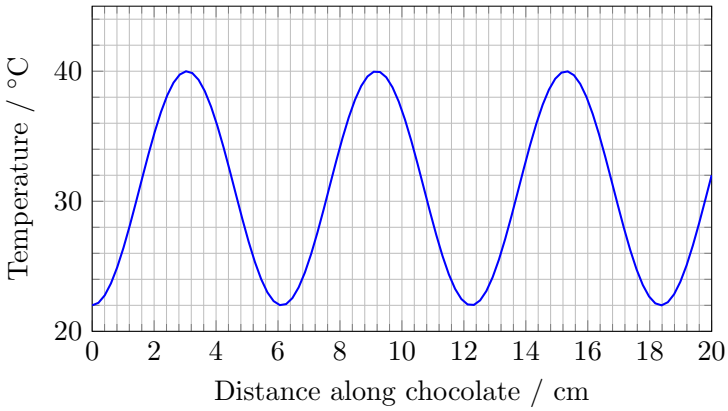
(i) State two necessary conditions for a stationary wave to form. [2]

(ii) Explain why the formation of a stationary wave inside the oven leads to uneven heating ("hot spots" and "cold spots") in the food. [2]

(d) Experimental Data Analysis

A student removes the rotating turntable from the microwave and places a flat, uniform layer of chocolate on a non-metallic tray. The microwave is turned on for a short time. The student observes that the chocolate melts in specific, evenly spaced spots. They measure the temperature of the chocolate along a straight line passing through several melted spots. The data is plotted in Figure 1.

Figure 1: Temperature profile of chocolate



(i) Use Figure 1 to determine the distance between adjacent antinodes in the stationary wave pattern. [1]

(ii) Use your answer to (d)(i) to calculate an experimental value for the speed of the microwaves, given the frequency is 2.45 GHz. Compare this to the accepted speed of light in a vacuum and suggest one reason for any minor discrepancy. [4]

(e) Thermal Physics

The microwave oven has a useful power output of 800 W delivered to the food. A student places a beaker containing 250 g of liquid water at 20°C into the oven.

(i) Calculate the theoretical time taken to heat the water to 95°C. Assume all the useful power output goes into heating the water and ignore the heat capacity of the beaker. [3]

(ii) In practice, the time taken is longer than the value calculated in (e)(i). Suggest one reason why. [1]

Mark Scheme: Physics of the Microwave Oven

1(a) The Magnetron (Electromagnetism)

(i) Thermionic emission **[1]**

(ii) Magnetic force provides the centripetal force:
$Bev = \frac{m_e v^2}{r}$ **[1]**
Rearranging for velocity: $v = \frac{Ber}{m_e}$
Substitute $v = \omega r = 2\pi f r$:
$2\pi f r = \frac{Ber}{m_e}$ **[1]**
Cancel r and rearrange for f to show it is independent of velocity:
$f = \frac{Be}{2\pi m_e}$ **[1]**

(iii) Rearrange equation for B: $B = \frac{2\pi m_e f}{e}$
$B = \frac{2\pi \times (9.11\times10^{-31}) \times (2.45\times10^{9})}{1.60\times10^{-19}}$ **[1]**
$B = 0.0877$ T (or 87.7 mT) **[1]**

1(b) Heating Mechanism (Electric Fields & Matter)

(i) A molecule with a permanent, uneven distribution of electric charge (one end is slightly positive, the other slightly negative). **[1]**

(ii) The alternating/oscillating electric field exerts a varying force on the dipoles, causing them to constantly rotate/flip to align with the field. **[1]**
This rapid rotation causes friction/collisions with neighboring molecules. **[1]**
These collisions transfer kinetic energy, increasing the internal energy (and therefore the temperature) of the food. **[1]**

1(c) Standing Waves (Wave Mechanics)

(i) Two waves traveling in opposite directions (must mention opposite directions). **[1]**
The waves must have the same frequency/wavelength (and similar amplitude). **[1]**

(ii) Nodes are points of zero amplitude/intensity, meaning no energy is transferred to the water molecules here, creating "cold spots". **[1]**
Antinodes are points of maximum amplitude/intensity, meaning maximum energy transfer and heating, creating "hot spots". **[1]**

1(d) Experimental Data Analysis

(i) 6.1 cm to 6.2 cm (Accept precise reading of 6.12 cm based on the mathematical model). **[1]**

(ii) The distance between adjacent antinodes is half a wavelength ($\frac{\lambda}{2}$).
$\lambda = 2 \times 6.12$ cm $= 12.24$ cm $= 0.1224$ m **[1]**
$v = f\lambda = (2.45 \times 10^9) \times 0.1224 = 2.9988 \times 10^8$ m s^{-1} **[1]**
Comparison: This is approximately 3.00×10^8 m s^{-1}, which matches the accepted speed of electromagnetic waves in a vacuum. **[1]**
Discrepancy reason (Any one of): Uncertainty in measuring the exact center of the melted spots / the speed of microwaves in air/chocolate is very slightly slower than in a perfect vacuum. **[1]**

1(e) Thermal Physics

(i) Calculate energy required: $Q = mc\Delta\theta$
$Q = 0.250 \times 4180 \times (95 - 20)$ **[1]**
$Q = 78375$ J **[1]**
Calculate time: $t = \frac{Q}{P} = \frac{78375}{800} = 98.0$ s **[1]**

(ii) Any sensible reason, such as: Energy is lost to the surroundings / Some energy is used to heat the beaker itself / Evaporation of water removes latent heat, requiring more total energy input. **[1]**

A-Level Physics (CIE 9702) Paper 4 Style

Synoptic Structured Question: Microwave Oven Physics

Variant 2 (Higher Difficulty)

Data

- Speed of electromagnetic waves in vacuum, $c = 3.00 \times 10^{8}\,\mathrm{m\,s^{-1}}$
- Elementary charge, $e = 1.60 \times 10^{-19}\,\mathrm{C}$
- Electron mass, $m_e = 9.11 \times 10^{-31}\,\mathrm{kg}$
- Specific heat capacity of water, $c_w = 4180\,\mathrm{J\,kg^{-1}\,K^{-1}}$
- Permittivity of free space, $\varepsilon_0 = 8.85 \times 10^{-12}\,\mathrm{F\,m^{-1}}$

Comprehension Passage (read before answering)

A domestic microwave oven uses a magnetron to generate electromagnetic radiation of nominal frequency 2.45 GHz. Inside the cavity, reflected waves can form standing-wave patterns, so energy transfer to food is spatially non-uniform. In one observation, adjacent melted spots on chocolate are separated by about 6.0 cm. The oven useful output power is approximately 800 W. A heating test uses 250 g of water, initially at 20 °C, and compares theoretical heating to measured outcomes. **Answer all parts.** The nominal microwave frequency is 2.45 GHz unless otherwise stated.

(a) A magnetron is operated so that electrons move in circular paths under a uniform magnetic flux density B.

(i) Show that the cyclotron angular frequency ω is given by $\omega = \dfrac{eB}{m_e}$ and deduce the linear frequency expression. [3]

(ii) The oven is designed for emission at 2.45 GHz. Calculate the required magnetic flux density. [2]

(iii) Open-ended: Explain, using physics principles, why an increase in the accelerating potential of electrons does not directly increase the microwave frequency of the cavity mode in a correctly tuned oven. [3]

(b) A standing-wave pattern forms in one dimension across the oven cavity.

(i) A row of melted spots in chocolate is separated by 6.0 cm. Determine the wavelength and calculate the wave speed. [3]

(ii) Open-ended: The measured speed is 2.5% below c. Discuss two physically valid reasons for this discrepancy and explain which is likely to be dominant in a school-laboratory setup. [4]

(c) A beaker contains 250 g of water initially at 20 °C. The microwave useful output is 800 W.

(i) Calculate the ideal time to raise the water temperature to 95 °C. [3]

(ii) In a repeat run, the measured time is 125 s. Determine the overall heating efficiency. [2]

(iii) Open-ended: Suggest and justify one experimental modification that would improve the reliability of the measured efficiency, and one that would improve its validity. [4]

(d) The electric field in part of the cavity is modeled by

$$E(x,t) = E_0 \sin(kx)\cos(\omega t).$$

(i) State the positions of nodes and antinodes in terms of k. [2]

(ii) Show that the time-averaged energy density is

$$\langle u \rangle = \frac{1}{4}\varepsilon_0 E_0^2 \sin^2(kx).$$

[2]

(iii) Open-ended: Explain why a rotating turntable reduces macroscopic temperature non-uniformity even though it does not remove the standing-wave field pattern itself. [2]

[Total: 30 marks]

Sample Solution and Mark Scheme

CIE 9702 Paper 4 Style — Variant 2

(a) Magnetron dynamics

(i) From magnetic force = centripetal force,

$$Bev = \frac{m_e v^2}{r}.$$

So $v/r = eB/m_e = \omega$ and

$$f = \frac{\omega}{2\pi} = \frac{eB}{2\pi m_e}.$$

M1 magnetic = centripetal, A1 for $\omega = eB/m_e$, A1 for frequency form. [3]

(ii)

$$B = \frac{2\pi m_e f}{e} = \frac{2\pi(9.11 \times 10^{-31})(2.45 \times 10^9)}{1.60 \times 10^{-19}} = 8.76 \times 10^{-2}\,\mathrm{T}.$$

$B \approx 0.088\,\mathrm{T}$ (88 mT). M1 substitution, A1 correct value/unit. [2]

(iii) Any coherent explanation including: microwave frequency set mainly by resonant cavity dimensions/mode structure (or by cyclotron condition when tuned), not simply electron kinetic energy; raising accelerating potential mainly changes electron speed/current/power transfer and interaction efficiency; frequency remains near cavity resonance, with only small pulling effects. B1 + B1 + B1. [3]

(b) Standing waves and measured speed

(i) Adjacent antinodes separated by $\lambda/2$, so

$$\lambda = 2(6.0\,\mathrm{cm}) = 12.0\,\mathrm{cm} = 0.120\,\mathrm{m}.$$

$$v = f\lambda = (2.45 \times 10^9)(0.120) = 2.94 \times 10^8\,\mathrm{m\,s^{-1}}.$$

M1 for $\lambda = 2d$, A1 wavelength, A1 speed. [3]

(ii) Two valid reasons, e.g.: non-vacuum medium effects (air, water vapour, food dielectric loading) reducing phase speed; position uncertainty of melt centres / thermal diffusion broadening spots; imperfect one-dimensional assumption / multimode field superposition. Plus justified dominance: measurement-position uncertainty and thermal spread usually dominate in school experiments. B1 + B1 + B1 (justification quality) + B1 (dominant reason argued). [4]

(c) Heating and efficiency

(i)

$$Q = mc\Delta T = (0.250)(4180)(95 - 20) = 78\,375\,\mathrm{J}.$$

$$t = Q/P = 78375/800 = 97.97\,\mathrm{s} \approx 98\,\mathrm{s}.$$

M1 for Q, A1 energy, A1 time. [3]

(ii) Useful power to water from observation:

$$P_{\mathrm{obs}} = Q/t = 78375/125 = 627\,\mathrm{W}.$$

$$\eta = P_{\mathrm{obs}}/800 = 0.784 \approx 78\%.$$

M1 observed power or equivalent ratio, A1 efficiency. [2]

(iii) Reliability improvement (one, with reason): repeat trials and average; use same starting mass and temperature; stirred temperature measurement. Validity improvement (one, with reason): insulated/lidded beaker to reduce non-target losses; calibrated probe at defined depth; account for beaker heat capacity. Award up to 2 marks each: one for practical change, one for justification. B2 + B2. [4]

(d) Field model and interpretation

(i) Nodes where $\sin(kx) = 0 \Rightarrow kx = n\pi \Rightarrow x = n\pi/k$. Antinodes where $|\sin(kx)| = 1 \Rightarrow kx = (2n+1)\pi/2 \Rightarrow x = (2n+1)\pi/(2k)$. A1 + A1. [2]

(ii) Instantaneous electric energy density

$$u_E = \frac{1}{2}\varepsilon_0 E^2 = \frac{1}{2}\varepsilon_0 E_0^2 \sin^2(kx)\cos^2(\omega t).$$

Using $\langle \cos^2(\omega t)\rangle = 1/2$,

$$\langle u \rangle = \frac{1}{4}\varepsilon_0 E_0^2 \sin^2(kx).$$

M1 setup with squared field, A1 averaged form. [2]

(iii) Turntable moves food through node/antinode regions over time, so each part samples different field intensities and receives more even time-averaged energy; standing-wave geometry in cavity remains unchanged because boundary conditions are unchanged. B1 + B1. [2]

Total: 30 marks